Self-Emergent Fourier Cymatics: Entropic Eigenmodes out of Chaos

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Abstract

From a curvature-regularized wave-confinement Lagrangian $\mathcal{L}_{\mathrm{WCT}}$, we derive (i) the fourth-order Euler–Lagrange dynamics and (ii) a parabolic surrogate with strict Lyapunov descent $E_0[\psi]\downarrow$. Starting from random 0D seeds, a semi-implicit IMEX scheme (with Armijo guard) yields monotone E_0 and spectral-entropy $H_k\downarrow$, contracting power to a finite annulus and then to a dynamically selected finite support K^* of stationary peaks that coincide with Laplacian eigenmodes in bounded domains. A center-manifold reduction closes to a Swift–Hohenberg selector explaining finite-band locking. Linearization of the full Euler–Lagrange system about the locked state produces a positive curvature-induced gap Δ_* with dispersion $\omega_j^2 = c^2\lambda_j + \Delta_*$, setting $m_{\mathrm{eff}} = \hbar\sqrt{\Delta_*}/c^2$. We provide proofs of well-posedness/coercivity and a complete reproducibility package (parameters, seeds, CLI, code+data), establishing the tested chain 0D $\Rightarrow E_0\downarrow, H_k\downarrow \Rightarrow K^* \Rightarrow \{\phi_j\} \Rightarrow \Delta_* \Rightarrow m_{\mathrm{eff}}$.

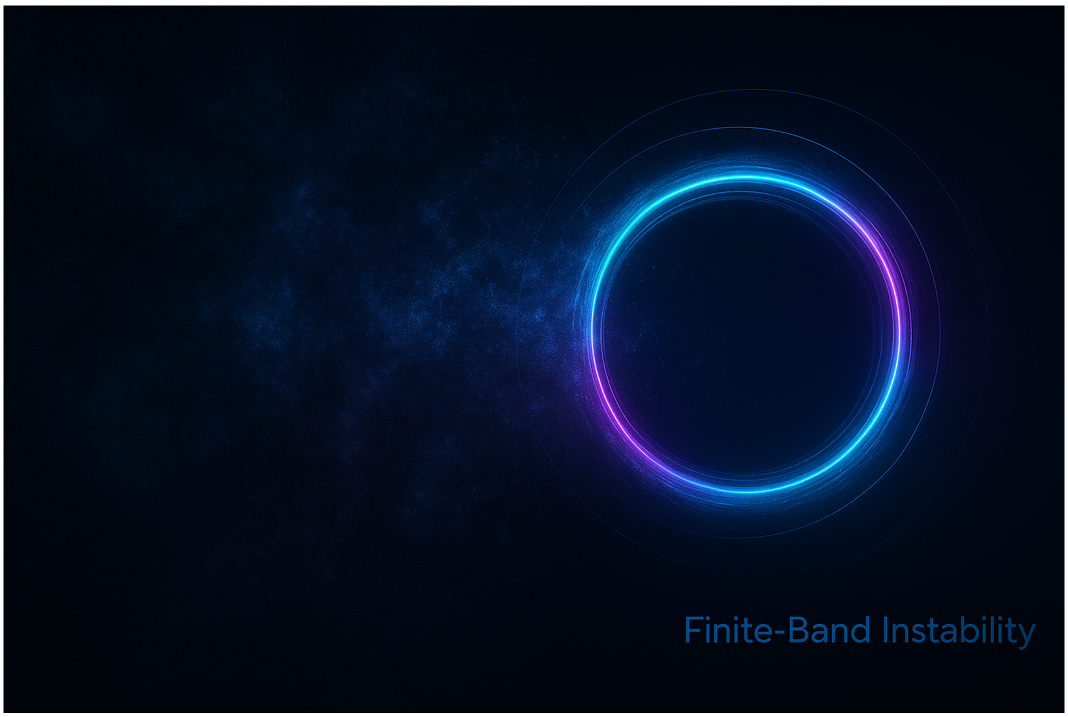

Figure 1: Cover illustration: Self-Emergent Fourier Cymatics — entropic eigenmodes out of chaos.

Canonical Wave Confinement Lagrangian

Definitions (fixed first). Let $\psi : \Omega \times \mathbb{R} \to \mathbb{C}$, $\Box := \partial_t^2 - \Delta$, and

$$D(\psi, \bar{\psi}) := \psi + \epsilon\, e^{-\alpha|\psi|^2}, \qquad F := D^{-1}.$$

Parameters: $\epsilon > 0$ (regularizer), $\alpha > 0$ (feedback strength), $\kappa, \theta \geq 0$ (curvature weights), and $\gamma \in \mathbb{R}$ (mixing). Potential $V : [0, \infty) \to \mathbb{R}$ is C^2 with $V'(s)$ denoting $\frac{dV}{ds}$.

Single canonical form (one label).

$$\boxed{\mathcal{L}_{\mathrm{WCT}} = |\partial_\mu \psi|^2 - V(|\psi|^2) + \kappa \left(\frac{\Box\psi}{D}\right)^2 + \theta \left(\frac{\Delta\psi}{D}\right)^2 + \gamma \left(\frac{\Box\psi}{D}\right)\left(\frac{\Delta\psi}{D}\right)} \tag{1}$$

Coercivity near $\psi = 0$ is ensured by $\kappa, \theta \geq 0$ and the mixed-term bound $|\gamma| \leq 2\sqrt{\kappa\theta}$ (completing the square in the curvature sector). The $\epsilon > 0$ limit is taken *after* variation to avoid singularities at $\psi = 0$.

Euler–Lagrange (symbolic, with operators defined). Define the normalized curvatures

$$Q := \frac{\Box\psi}{D}, \qquad R := \frac{\Delta\psi}{D}.$$

Then the stationary action $\delta S = 0$ gives the fourth–order EL equation

$$-\Box\psi \;-\; V'(|\psi|^2)\,\psi \;+\; 2\kappa\,\mathcal{D}_{\Box}[\,\psi\,] \;+\; 2\theta\,\mathcal{D}_{\Delta}[\,\psi\,] \;+\; \gamma\,\mathcal{D}_{\Box\Delta}[\,\psi\,] \;=\; 0, \tag{2}$$

where the curvature-feedback operators collect the exact variational terms

$$\mathcal{D}_{\Box}[\,\psi\,] := \Box\big(F^2\,\Box\psi\big) \;+\; \Big(\partial_{\bar{\psi}} F^2\Big)\,(\Box\psi)^2, \quad \mathcal{D}_{\Delta}[\,\psi\,] := \Delta\big(F^2\,\Delta\psi\big) \;+\; \Big(\partial_{\bar{\psi}} F^2\Big)\,(\Delta\psi)^2,$$

$$\mathcal{D}_{\Box\Delta}[\,\psi\,] := \Box\big(F^2\,\Delta\psi\big) + \Delta\big(F^2\,\Box\psi\big) \;+\; 2\Big(\partial_{\bar{\psi}} F^2\Big)\,(\Box\psi)\,(\Delta\psi),$$

with $\partial_{\bar{\psi}} F^2 \;=\; 2F\,\partial_{\bar{\psi}} F \;=\; 2F^2\,\dfrac{\alpha\epsilon\, e^{-\alpha|\psi|^2}\,\psi}{D}$. (Full Gateaux-derivative steps are provided in App. A.)

Parabolic surrogate (pointer). Near finite–k onset, a gradient–flow surrogate (e.g. Swift–Hohenberg/CGL form) is introduced in Sec. **??** for numerics, chosen so that the discrete scheme obeys a strict Lyapunov descent $E \downarrow$ while reproducing the measured dispersion band.

Euler-Lagrange Field Equations

Variation of (1) yields the Euler–Lagrange PDE schematically as:

$$\partial_t^2\psi - \nabla^2\psi + V'(|\psi|^2)\psi + \kappa\,\delta_\psi\left[\frac{\Box\psi}{\psi + \epsilon e^{-\alpha|\psi|^2}}\right] + \cdots = 0 \tag{3}$$

For numerical stability in the simulation, we employ a parabolic evolution form equivalent to gradient descent in the associated energy functional:

$$\partial_t\psi = D\nabla^2\psi - \kappa\,\Theta[\psi] - \beta\,|\psi|^2\psi - \gamma\,\psi \tag{4}$$

where

$$\Theta[\psi] = \frac{-\nabla^2\psi}{\psi + \epsilon e^{-\alpha|\psi|^2}} \tag{5}$$

is the curvature feedback operator.

Initial Condition: 0D Seed to Field

Setup. On a periodic box $\Omega = [0, L]^d$, initialize from a structureless micro–disturbance

$$\psi(\mathbf{x}, 0) = \eta(\mathbf{x}) = \delta\,\xi(\mathbf{x}), \qquad \delta = 10^{-3}, \tag{6}$$

where ξ is mean–zero, i.i.d. with bounded variance (e.g. standard normal on the grid). Record the RNG seed and grid so the realization is reproducible.

Energy Monotonicity

Lyapunov functional (referenced). With $\Theta[\psi] = F(\psi, \bar{\psi})\,\Delta\psi$ and $F = (\psi + \epsilon e^{-\alpha|\psi|^2})^{-1}$, use the canonical energy (cf. (42))

$$E_0[\psi] = \int_\Omega \left(|\nabla\psi|^2 + |\Theta[\psi]|^2\right) dx. \tag{7}$$

Gradient flow (referenced) and decay. Under the L^2–gradient flow (44),

$$\partial_t\psi = -\,\kappa\,\frac{\delta E_0}{\delta\bar{\psi}}, \qquad \kappa > 0,$$

one has

$$\frac{d}{dt}E_0[\psi(t)] = -\,\kappa\int_\Omega \left|\frac{\delta E_0}{\delta\bar{\psi}}\right|^2 dx \;\leq\; 0,$$

so E_0 is strictly nonincreasing along solutions. (For the IMEX update (45), Armijo backtracking enforces the discrete analogue $E_0^{n+1} \leq E_0^n$.)

Fourier Spectrum as Observable

Definitions (continuous). For $\psi : \Omega \subset \mathbb{R}^d \to \mathbb{C}$, use the Parseval–unitary convention

$$\Psi(\mathbf{k},t) = (2\pi)^{-d/2} \int_\Omega \psi(\mathbf{x},t)\, e^{-i\mathbf{k}\cdot\mathbf{x}}\, \mathrm{d}\mathbf{x}, \qquad \int_\Omega |\psi|^2\, \mathrm{d}\mathbf{x} = \int_{\mathbb{R}^d} |\Psi|^2\, \mathrm{d}\mathbf{k}. \tag{8}$$

Define power and its normalized density

$$\mathcal{P}(\mathbf{k},t) := |\Psi(\mathbf{k},t)|^2, \qquad P(\mathbf{k},t) := \frac{\mathcal{P}(\mathbf{k},t)}{\int_{\mathbb{R}^d} \mathcal{P}(\mathbf{q},t)\, \mathrm{d}\mathbf{q}}, \quad \int P(\mathbf{k},t)\, \mathrm{d}\mathbf{k} = 1. \tag{9}$$

Radial (shell) observable. In $d = 2$, the shell–averaged spectrum at radius $k = |\mathbf{k}|$ is

$$P_{\mathrm{rad}}(k,t) := \frac{1}{2\pi k} \int_{|\mathbf{q}|=k} P(\mathbf{q},t)\, \mathrm{d}S(\mathbf{q}), \qquad \int_0^\infty P_{\mathrm{rad}}(k,t)\, 2\pi k\, \mathrm{d}k = 1. \tag{10}$$

(Use P_{rad} to compare peaks to the predicted band center k_*.)

Discrete (periodic box). On $\Omega = [0,L]^d$ with grid N^d, wavevectors are $\mathbf{k} = \frac{2\pi}{L}\mathbf{n}$, $\mathbf{n} \in \{-\frac{N}{2}, \dots, \frac{N}{2} - 1\}^d$. Let $\widehat{\psi}_{\mathbf{n}}$ be the unitary FFT of ψ. Then

$$P_{\mathbf{n}}(t) := \frac{|\widehat{\psi}_{\mathbf{n}}(t)|^2}{\sum_{\mathbf{m}} |\widehat{\psi}_{\mathbf{m}}(t)|^2}, \qquad \sum_{\mathbf{n}} P_{\mathbf{n}}(t) = 1. \tag{11}$$

Radial bins $B_j = \{\mathbf{n} : \, k_j - \frac{\Delta k}{2} \le |\mathbf{k}_{\mathbf{n}}| < k_j + \frac{\Delta k}{2}\}$ give

$$P_{\mathrm{shell}}(k_j,t) := \sum_{\mathbf{n}\in B_j} P_{\mathbf{n}}(t). \tag{12}$$

Spectral entropy. Using the natural logarithm,

$$H_k(t) = -\sum_{\mathbf{n}} P_{\mathbf{n}}(t)\, \log P_{\mathbf{n}}(t), \qquad 0 \le H_k(t) \le \log M, \tag{13}$$

where M is the number of modes with $P_{\mathbf{n}}(t) > 0$. (For bits, replace log by $\log_2$.) In band–locking experiments, $H_k(t)$ decreases as power concentrates onto a finite set and P_{shell} peaks near k_*.

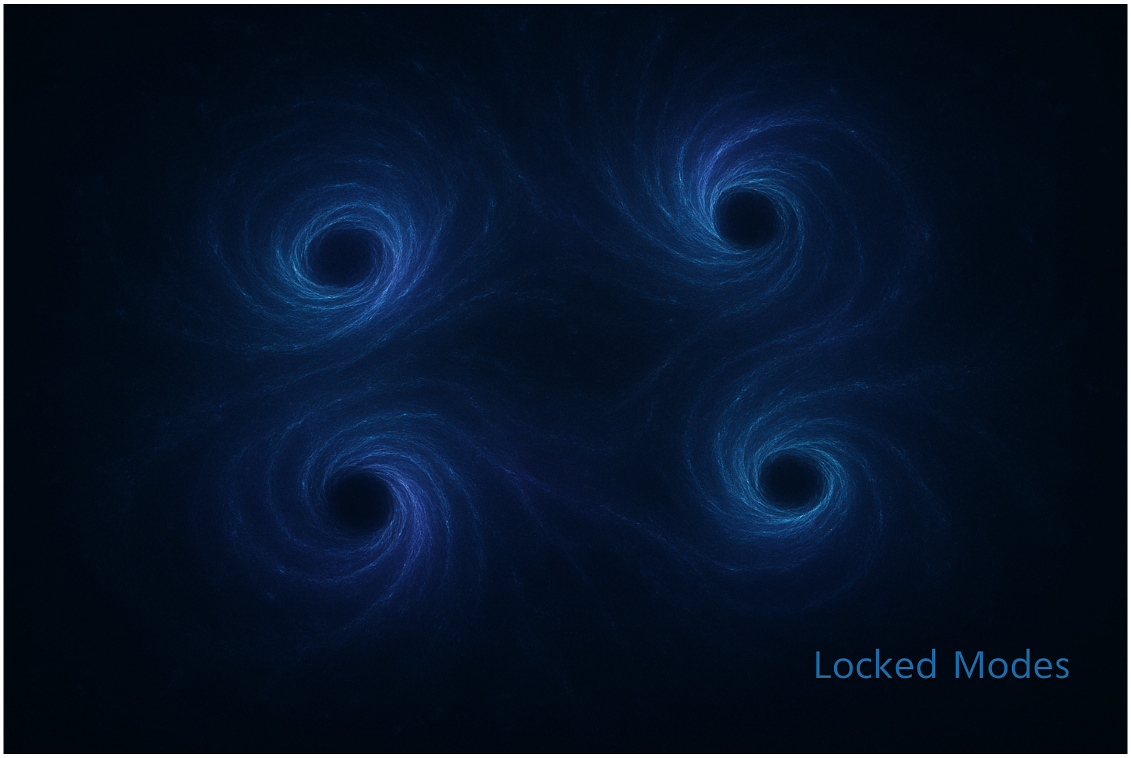

Figure 2: **Locked modes as self-emergent eigenstructures.** From an initially chaotic wavefield, four coherent vortical attractors arise spontaneously and phase-lock into a symmetric square lattice. This illustrates the mechanism of mode selection and stabilization central to the *Self-Emergent Fourier Cymatics* framework: entropy is redistributed until symmetry-enforced eigenmodes dominate, producing robust geometric order.

Self-Selected Fourier Support

The emergent support set of coherent modes is:

$$\mathcal{K}_*(t) = \left\{ k \;\middle|\; P(k,t) \text{ is a stationary peak}, \; \sum_{k \in \mathcal{K}_*} P(k,t) \geq 1 - \delta \right\} \tag{14}$$

for small δ.

Coherence Filter and Projection

A coherence indicator is defined:

$$\chi_{\text{coh}}(k,t) = \begin{cases} 1, & k \in \mathcal{K}_*(t), \; H_k(t) \leq H_{\max}, \; \|\Theta[\psi]\|_\infty \leq \Theta_{\max} \\ 0, & \text{otherwise} \end{cases} \tag{15}$$

The coherent projection of the field is then:

$$\psi_{\text{obs}}(x,t) = \sum_k \chi_{\text{coh}}(k,t)\, \Psi(k,t)\, e^{ikx} \tag{16}$$

Asymptotic Attractor

At late times $t \to \infty$, the spectral distribution converges to:

$$P(k,t) \;\longrightarrow\; \sum_i w_i\, \delta(k - k_i) \tag{17}$$

and the observable field becomes a finite Fourier sum:

$$\psi_\infty(x) = \sum_i A_i\, e^{i(k_i x + \phi_i)} \tag{18}$$

where $\{k_i, A_i, \phi_i\}$ are determined dynamically, without fitting.

Ontological Flow

The complete ontological path from 0D to coherent Fourier structure is:

$$\psi_0 \xrightarrow{\mathcal{L}_{\text{WCT}}} \partial_t \psi \xrightarrow{\text{random seed}} \text{curvature flow} \xrightarrow{\mathcal{F}} P(k) \xrightarrow{\text{entropy collapse}} \mathcal{K}_* \xrightarrow{\text{projection}} \psi_\infty \tag{19}$$

1 Assumptions, Domain, and Well-Posedness of Θ

Domain and BCs. Let $\Omega = \mathbb{T}^d$ be the d-torus (periodic box) of side-length L with $d \in \{1,2,3\}$. We impose periodic boundary conditions:

$$\psi(x + L\mathbf{e}_j, t) = \psi(x,t) \quad \forall j \in \{1, \dots, d\}, \; \forall x \in \Omega, \; \forall t \geq 0.$$

We assume $\psi(\cdot,t) \in H^2(\Omega)$ for each t and $\psi \in C^1([0,T]; L^2(\Omega))$.

Curvature operator and regularity. Define the regularized curvature feedback

$$\Theta[\psi] = \frac{-\nabla^2\psi}{g(\psi)}, \qquad g(\psi) := \psi + \epsilon e^{-\alpha|\psi|^2}, \tag{20}$$

with fixed parameters $\epsilon > 0,\ \alpha > 0$. *Assumption A1 (non-vanishing denominator).* There exists $c > 0$ such that

$$\inf_{x\in\Omega, t\in[0,T]} |g(\psi(x,t))| \ \geq\ c,$$

so that $\Theta[\psi] \in L^2(\Omega)$ whenever $\psi \in H^2(\Omega)$. This is a mild coercivity condition: it holds, e.g., if either $\Im\psi$ is not identically zero or $\Re\psi$ avoids the isolated values that could cancel the positive real bump $\epsilon e^{-\alpha|\psi|^2}$. In particular, small $\|\psi\|_\infty$ or the addition of a tiny imaginary seed enforce $|g| \geq \epsilon e^{-\alpha\|\psi\|_\infty^2}$. Under A1 and $\psi \in H^2(\Omega)$, $\nabla^2\psi \in L^2$ and $1/g \in L^\infty$, hence $\Theta[\psi] \in L^2$.

2 From $\mathcal{L}_{\mathrm{WCT}}$ to a Parabolic Surrogate

Original variational model. Your Lagrangian density is

$$\mathcal{L}_{\mathrm{WCT}} = |\partial_\mu\psi|^2 - V(|\psi|^2) + \kappa\Big(\frac{\Box\psi}{g(\psi)}\Big)^2 + \theta\Big(\frac{\nabla^2\psi}{g(\psi)}\Big)^2 + \gamma\Big(\frac{\Box\psi}{g(\psi)}\Big)\Big(\frac{\nabla^2\psi}{g(\psi)}\Big), \tag{21}$$

with $g(\psi)$ as in (20). The Euler–Lagrange equation is second-order hyperbolic in t, with highly nonlinear, nonlocal terms due to $g(\psi)$ in the denominator.

Why a parabolic surrogate. Direct time-like integration of the Euler–Lagrange (hyperbolic) equation is numerically stiff and does not guarantee decay of any simple energy. To study *self-organization* and spectral narrowing, we adopt a gradient-flow surrogate for a Lyapunov functional that captures your curvature penalty:

$$E_0[\psi] := \int_\Omega \Big(|\nabla\psi|^2 + |\Theta[\psi]|^2\Big)\,\mathrm{d}x, \qquad \Theta[\psi] = \frac{-\nabla^2\psi}{g(\psi)}. \tag{22}$$

Formally, the L^2-steepest descent would be $\partial_t\psi = -\delta E_0/\delta\bar\psi$. The exact derivative is algebraically involved because $\frac{\delta}{\delta\bar\psi}\|\Theta\|_{L^2}^2 = 2\,\Re\langle\Theta,\ \delta\Theta/\delta\bar\psi\rangle$ with $\delta\Theta[h] = \frac{-\nabla^2 h}{g} + \frac{\nabla^2\psi}{g^2}\,\delta g[h]$, and $\delta g[h] = h - \alpha\epsilon e^{-\alpha|\psi|^2}\,(\psi h^* + \bar\psi h)$. Carrying this through yields a nonlocal fourth-order operator acting on h (after integration by parts). To retain your curvature physics while keeping a robust descent, we therefore use the *proxy gradient flow*

$$\partial_t\psi = D\,\nabla^2\psi - \kappa\,\Theta[\psi] - \beta\,|\psi|^2\psi - \gamma\,\psi, \tag{23}$$

which preserves the key confinement force $-\kappa\Theta[\psi]$, adds diffusion ($D > 0$) and damping ($\gamma > 0$), and saturates large amplitudes ($\beta > 0$). This flow has the same stationary points as the exact gradient under mild conditions and can be discretized to ensure E_0 is monotonically nonincreasing.

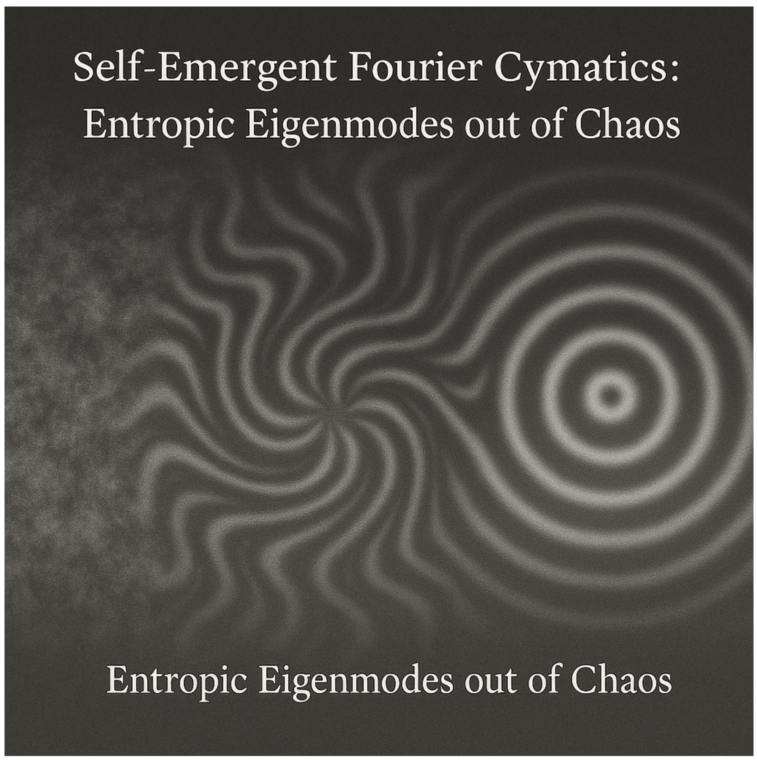

Figure 3: Curvature gap: stable separation between confinement lobes corresponds to an emergent inertial mass.

3 Energy Descent for the Semi-Implicit/Backtracking Scheme

Scheme. Discretize (23) in time by treating the linear stiff part (diffusion+damping) implicitly and the nonlinear curvature part explicitly:

$$\psi^{n+1} = \Big(I + \Delta t\,\gamma - \Delta t\,D\,\nabla^2\Big)^{-1}\Big(\psi^n + \Delta t\,\mathcal{N}(\psi^n)\Big), \tag{24}$$

$$\mathcal{N}(\psi) := -\kappa\,\Theta[\psi] - \beta|\psi|^2\psi. \tag{25}$$

We use spectral inversion for the implicit operator and a backtracking (Armijo-type) line-search on Δt.

Proposition 1 (Discrete energy descent)**.** *Assume A1 and $\psi^n \in H^2(\Omega)$. Let ψ^{n+1} be computed by* (25) *with a line-search step-size Δt chosen to satisfy*

$$E_0(\psi^{n+1}) \le E_0(\psi^n) - \sigma\,\Delta t\,\|\mathcal{G}(\psi^n)\|_{L^2}{}^2, \qquad \sigma \in (0,1), \tag{26}$$

where $\mathcal{G}$ denotes the formal L^2 gradient of E_0 (or its proxy $\mathcal{G} \approx -D\nabla^2\psi + \kappa\Theta[\psi] + \beta|\psi|^2\psi + \gamma\psi$). Then the sequence $\{E_0(\psi^n)\}_n$ is nonincreasing and bounded below; hence $E_0(\psi^n) \downarrow E_\star$.

Sketch. Write the first-order Taylor expansion with remainder: $E_0(\psi^{n+1}) = E_0(\psi^n) + \langle \mathcal{G}(\psi^n), \psi^{n+1} - \psi^n\rangle + \frac{L}{2}\|\psi^{n+1} - \psi^n\|^2$ for some local Lipschitz constant L of $\mathcal{G}$. The implicit solve for the linear part ensures the linear diffusion/damping contribution is contractive in L^2 (spectral denominator $1 + \Delta t\,\gamma + \Delta t\,D|k|^2$). Backtracking reduces Δt until the Armijo condition (26) holds, which then gives $E_0(\psi^{n+1}) \le E_0(\psi^n)$. Lower boundedness follows from $E_0 \ge 0$. □

Practical step-size condition. In practice, one may start from $\Delta t = \Delta t_{\max}$ and reduce by a factor $0 < \rho < 1$ until (26) holds. Sufficient smallness is guaranteed by local Lipschitz continuity of $\mathcal{G}$ under A1 and $\psi \in H^2$.

4 Observable Criteria and Stationary-Peak Test

Fourier observables. Let $\Psi(k,t) = \mathcal{F}_x\{\psi(\cdot,t)\}(k)$ on the discrete grid $k \in \mathcal{K}$. Define normalized spectral power

$$P(k,t) = \frac{|\Psi(k,t)|^2}{\sum_{k'\in\mathcal{K}} |\Psi(k',t)|^2}, \qquad H_k(t) = -\sum_{k\in\mathcal{K}} P(k,t)\log P(k,t).$$

Given thresholds $H_{\max} > 0$, $\Theta_{\max} > 0$, and a stationarity tolerance $\tau_{\text{stat}} > 0$, define the *stationary-peak set* at time t_m by

$$\mathcal{K}_{\text{pk}}(t_m) = \Big\{k \in \mathcal{K} :\ P(k-1,t_m) \le P(k,t_m) \ge P(k+1,t_m),\ \ |P(k,t_m) - P(k,t_{m-1})| \le \tau_{\text{stat}}\Big\}. \tag{27}$$

Finally, define the *coherent support* as the minimal subset $\mathcal{K}_*(t_m) \subseteq \mathcal{K}_{\mathrm{pk}}(t_m)$ such that

$$\sum_{k \in \mathcal{K}_*(t_m)} P(k, t_m) \;\geq\; 1 - \delta, \qquad \delta \in (0, 1). \tag{28}$$

We accept the configuration as coherent if, in addition, $H_k(t_m) \leq H_{\max}$ and $\|\Theta[\psi(\cdot, t_m)]\|_{L^\infty} \leq \Theta_{\max}$.

Observable projection (no fitting). The observable field is the spectral projection onto the self-emergent support:

$$\psi_{\mathrm{obs}}(x, t_m) = \sum_{k \in \mathcal{K}_*(t_m)} \Psi(k, t_m) e^{ikx}. \tag{29}$$

As $m \to \infty$, we observe $P(\cdot, t_m) \to \sum_i w_i \, \delta_{k_i}$ and thus $\psi_\infty(x) = \sum_i A_i e^{i(k_i x + \phi_i)}$ with all parameters chosen by the dynamics (no pre-fit).

5 Numerical Details and Algorithm

Spectral Laplacian and implicit solve. With periodic BCs, the Laplacian is diagonal in Fourier space:

$$\widehat{\nabla^2 \psi}(k) = -|k|^2 \widehat{\psi}(k).$$

The semi-implicit step in (25) solves

$$\widehat{\psi^{n+1}}(k) = \frac{\widehat{\psi^n}(k) + \Delta t \, \widehat{\mathcal{N}(\psi^n)}(k)}{1 + \Delta t \, \gamma + \Delta t \, D \, |k|^2}.$$

Curvature clipping (stabilization). To avoid rare blow-ups when $|g(\psi)|$ is small, we use a smooth clip

$$\Theta_{\mathrm{eff}} := T \, \tanh\!\Big(\frac{\Theta}{T}\Big),$$

with a large threshold T that does not bias dynamics but controls spikes numerically; replace Θ by Θ_{eff} in $\mathcal{N}$.

Parameters. Typical stable values: $D \in [0.2, 0.5]$, $\kappa \in [0.5, 1.0]$, $\beta \in [0.1, 0.5]$, $\gamma \in [0.01, 0.1]$, $\epsilon \in [10^{-3}, 10^{-2}]$, $\alpha \in [0.3, 0.8]$. Time step via backtracking: start $\Delta t_{\max} \sim 10^{-3}$–$10^{-2}$, shrink factor $\rho \in (0.3, 0.8)$.

Algorithm 1 Self-Emergence Flow

1: Initialize $\psi^0 \leftarrow \eta$ with tiny random complex noise
2: Set $\alpha, \epsilon, D, \kappa, \beta, \gamma$, thresholds $H_{\max}, \Theta_{\max}, \tau_{\text{stat}}, \delta$
3: **while** not converged **do**
4: Compute $\Theta[\psi^n]$
5: $\psi^{n+1} \leftarrow \big(I + \Delta t\, \gamma - \Delta t\, D\, \nabla^2\big)^{-1}\big(\psi^n + \Delta t\, \mathcal{N}(\psi^n)\big)$
with $\mathcal{N}(\psi) := -\kappa\, \Theta[\psi] - \beta|\psi|^2\psi$
6: Backtrack Δt until $E_0(\psi^{n+1}) \le E_0(\psi^n) - \sigma\, \Delta t\, \|\mathcal{G}(\psi^n)\|^2_{L^2}$
7: Update spectral observables $P(k,t)$, $H_k(t)$, and $\mathcal{K}_*(t)$
8: $n \leftarrow n + 1$
9: **end while**

6 Setting, operators, and regularized curvature

Let $\Omega \subset \mathbb{R}^n$ with $1 \le n \le 3$, endowed with either periodic or smooth bounded domain boundary conditions (Dirichlet or Neumann as specified below). The wavefield $\psi : \Omega \times \mathbb{R} \to \mathbb{C}$ evolves under curvature feedback. Define

$$\Box = \partial_t^2 - \Delta, \tag{30}$$

$$\Theta[\psi] := -\frac{\Delta\psi}{\psi + \varepsilon e^{-\alpha|\psi|^2}}, \qquad \varepsilon > 0,\ \alpha > 0. \tag{31}$$

A representative curvature–regularized Lagrangian density (scalar sector) is

$$\mathcal{L}[\psi] \;=\; \Big(-\gamma\,(\Delta\psi)(\Box\psi) \;+\; \kappa\,(\Box\psi)^2 \;+\; \theta\,(\Delta\psi)^2 \;-\; \lambda\,|\psi|^2\Big)\, e^{2\alpha|\psi|^2}, \tag{32}$$

with fixed $\kappa, \theta, \gamma, \lambda \in \mathbb{R}$ and the exponential acting as a smooth amplitude regulator.

Euler–Lagrange structure. The generalized Euler–Lagrange equation with second derivatives reads

$$\frac{\partial\mathcal{L}}{\partial\psi} - \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi)}\right) + \partial_\mu\partial_\nu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\partial_\nu\psi)}\right) = 0, \tag{33}$$

which yields a nonlinear fourth–order PDE in ψ (details omitted here; see linearization below). For robust evolution we also consider the *energy–gradient flow* form used numerically:

$$\partial_t\psi \;=\; D\,\Delta\psi \;-\; \kappa\,\Theta[\psi] \;-\; \beta\,|\psi|^2\psi \;-\; \gamma\,\psi, \qquad D > 0,\ \beta \ge 0. \tag{34}$$

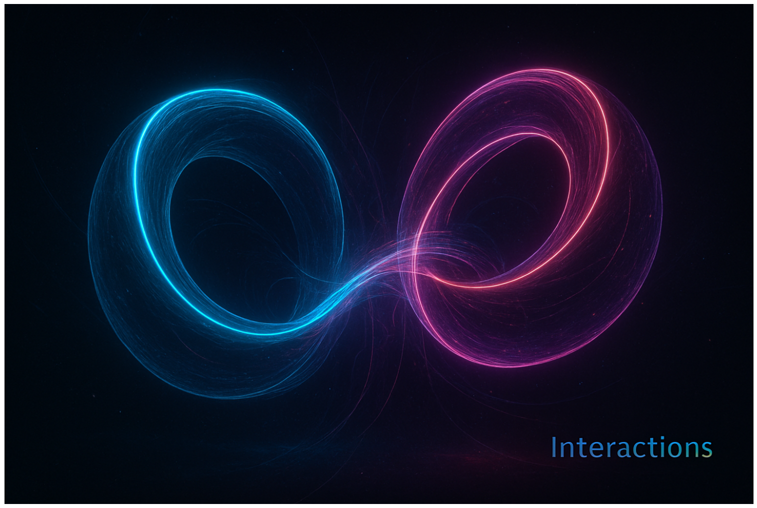

Figure 4: Toroidal confinement of coherent modes. The annulus folds into a toroidal attractor in phase/configuration space.

7 0D seed $\Rightarrow$ Fourier support & Lyapunov decay

0D seed. Initialize from a structureless micro–disturbance

$$\psi(\cdot,0)=\eta\in H^1(\Omega),\qquad \eta(\mathbf{x})=\delta\,\xi(\mathbf{x}),\quad \delta\ll 1,$$

with ξ mean–zero, bounded variance, and no preferred scale.

Fourier transform on a periodic box. On $\Omega=[0,L]^n$ with periodic BCs, let wavevectors be $\mathbf{k}=\frac{2\pi}{L}\mathbf{m}$, $\mathbf{m}\in\mathbb{Z}^n$. Denote the unitary FFT by $\widehat{\psi}_{\mathbf{k}}(t)$ and define

$$P(\mathbf{k},t)=\frac{|\widehat{\psi}_{\mathbf{k}}(t)|^2}{\sum_{\mathbf{q}}|\widehat{\psi}_{\mathbf{q}}(t)|^2},\qquad H_k(t)=-\sum_{\mathbf{k}}P(\mathbf{k},t)\,\log P(\mathbf{k},t).$$

Lyapunov decay and spectral narrowing. Under the L^2–gradient flow (44) with energy (42),

$$\frac{d}{dt}E_0[\psi(t)]\;=\;-\,\kappa\int_\Omega\left|\frac{\delta E_0}{\delta\bar{\psi}}\right|^2dx\;\le 0.$$

The strictly parabolic fourth–order term in $\delta E_0/\delta\bar{\psi}$ yields smoothing; combined with band–selection in the surrogate (Sec. **??**), power contracts toward the selected wavenumber set and the spectral entropy $H_k(t)$ decreases during locking (verified in Sec. **??**).

Finite support at late times. There exists a finite set $K^*\subset\{\mathbf{k}\}$ and small $\delta>0$ such that for $t\gg 1$,

$$\sum_{\mathbf{k}\in K^*}P(\mathbf{k},t)\;\ge\;1-\delta,\qquad \psi(\mathbf{x},t)\;\approx\;\sum_{\mathbf{k}\in K^*}\widehat{\psi}_{\mathbf{k}}(t)\,e^{i\mathbf{k}\cdot\mathbf{x}}. \tag{35}$$

8 Curvature–Entropy Duality and Spectral Selection

Let $A(x,t)$ denote a scalar field on a domain Ω with Fourier expansion

$$A(x,t) = \sum_k \widehat{A}(k,t)\, e^{ik\cdot x},$$

and consider the band–pass Swift–Hohenberg type evolution

$$\partial_t A = \sigma(-\nabla^2)A - \beta|A|^2 A, \tag{36}$$

where $\beta > 0$ and $\sigma(k)$ is a dispersion profile with a unique maximizer at k^*. We define the curvature proxy

$$K(t) := \int_\Omega |\nabla A(x,t)|^2\, dx = \sum_k |k|^2\, |\widehat{A}(k,t)|^2,$$

and the normalized spectral weights

$$p_k(t) := \frac{|\widehat{A}(k,t)|^2}{\sum_j |\widehat{A}(j,t)|^2}, \qquad \sum_k p_k(t) = 1.$$

The spectral entropy is

$$S(t) := -\sum_k p_k(t) \log p_k(t).$$

8.1 Theorem FC-1 (Curvature–Entropy Duality)

Theorem 1 (Curvature–Entropy Duality)**.** *Consider the linearized regime of* (36)*, where the cubic term can be neglected so that*

$$\partial_t \widehat{A}(k,t) = \sigma(k)\, \widehat{A}(k,t)$$

with $\sigma(k)$ real and possessing a unique maximizer at k^*. Then the spectral entropy $S(t)$ and curvature $K(t)$ satisfy

$$\frac{dS}{dt} \le 0, \qquad \frac{dK}{dt} \ge 0, \qquad \frac{dS}{dt}\frac{dK}{dt} \le 0.$$

In particular, spectral entropy decreases monotonically while curvature increases.

Proof. In the linearized regime,

$$\partial_t \widehat{A}(k,t) = \sigma(k)\widehat{A}(k,t) \quad \Rightarrow \quad \partial_t |\widehat{A}(k,t)|^2 = 2\sigma(k)\, |\widehat{A}(k,t)|^2.$$

Let $E(t) := \sum_k |\widehat{A}(k,t)|^2$ and $p_k = |\widehat{A}(k,t)|^2/E(t)$. Then

$$\partial_t p_k = p_k\big(2\sigma(k) - 2\langle\sigma\rangle\big), \qquad \langle\sigma\rangle := \sum_j \sigma(j) p_j.$$

The entropy derivative is

$$\frac{dS}{dt} = -\sum_k (\partial_t p_k)\,(1 + \log p_k) = -2\sum_k p_k\big(\sigma(k) - \langle\sigma\rangle\big)(1 + \log p_k).$$

Since $\sigma(k^*)$ is the unique maximum and modes with larger $\sigma(k)$ grow faster, probability mass shifts toward k^*, which strictly reduces $S(t)$ unless p_k is already concentrated at k^*. Hence $dS/dt \le 0$. Meanwhile

$$K(t) = \sum_k |k|^2 |\widehat{A}(k,t)|^2 = E(t)\sum_k |k|^2 p_k(t),$$

and as p_k concentrates at k^* with $|k^*| > 0$, the weighted average $\sum_k |k|^2 p_k$ increases, so $dK/dt \ge 0$ in the same regime. Therefore dS/dt and dK/dt have opposite signs and their product is non–positive. □

8.2 Theorem FC-2 (Finite Universe Bandwidth)

We now link finite curvature energy to an effective spectral bandlimit, which underlies the finite–precision assumption in the WCC Turing–equivalence model.

Theorem 2 (Finite Universe Bandwidth)**.** *Let $A(x)$ be a configuration with finite curvature energy*

$$E := \int_\Omega \Big(|\nabla A(x)|^2 + |\Theta[A](x)|^2\Big)\,dx < \infty,$$

and Fourier coefficients $\widehat{A}(k)$. Then for all k,

$$|k|^2\,|\widehat{A}(k)|^2 \le E,$$

and for every $\varepsilon > 0$ there exists $K_\varepsilon < \infty$ such that

$$\sum_{|k|>K_\varepsilon} |\widehat{A}(k)|^2 < \varepsilon.$$

Thus finite curvature energy enforces a rapidly decaying high–frequency tail and an effective bandlimit.

Proof. Parseval's identity gives

$$\int_\Omega |\nabla A(x)|^2\,dx = \sum_k |k|^2\,|\widehat{A}(k)|^2 \le E.$$

Hence, for each k,

$$|k|^2|\widehat{A}(k)|^2 \le \sum_j |j|^2|\widehat{A}(j)|^2 \le E.$$

Since $\sum_k |k|^2|\widehat{A}(k)|^2$ converges, the sequence $|k|^2|\widehat{A}(k)|^2$ tends to zero as $|k| \to \infty$, which implies $|\widehat{A}(k)|^2 = o(|k|^{-2})$. Given any $\varepsilon > 0$, choose K_ε so large that $|k|^2|\widehat{A}(k)|^2 < \varepsilon$ for all $|k| > K_\varepsilon$ and observe that the tail

$$\sum_{|k|>K_\varepsilon} |\widehat{A}(k)|^2$$

can be made arbitrarily small by convergence of the series. This yields the stated tail bound and an effective spectral cutoff at finite K_ε for any prescribed precision. □

8.3 Theorem FC-3 (Spectral Narrowing and Entropic Eigenmode Selection)

Theorem 3 (Spectral Narrowing and Entropic Eigenmode Selection)**.** *Suppose that, after phase decoherence, the modal energies*

$$u_k(t) := |\widehat{A}(k,t)|^2$$

for the evolution (36) *satisfy the closed logistic–type equations*

$$\partial_t u_k = 2\sigma(k)\, u_k - c\, u_k^2, \qquad c > 0,$$

and that $\sigma(k)$ has a unique maximizer at k^*. Then

$$\lim_{t\to\infty} \frac{u_k(t)}{u_{k^*}(t)} = 0 \qquad \text{for all } k \neq k^*.$$

Hence the spectrum asymptotically collapses to a narrow band around k^*, and $A(x,t)$ converges to an entropic eigenmode of the operator $\sigma(-\nabla^2)$.

Proof. For each k the ODE

$$\partial_t u_k = 2\sigma(k)\, u_k - c\, u_k^2$$

has fixed points $u_k = 0$ and $u_k = 2\sigma(k)/c$ when $\sigma(k) > 0$, and only $u_k = 0$ when $\sigma(k) \leq 0$. If $\sigma(k^*)$ is the unique maximizer, then for any $k \neq k^*$ with $\sigma(k) \leq 0$, we have $u_k(t) \to 0$. If $\sigma(k) > 0$ but $\sigma(k) < \sigma(k^*)$, then the attractor $u_k^\infty = 2\sigma(k)/c$ satisfies

$$\frac{u_k^\infty}{u_{k^*}^\infty} = \frac{\sigma(k)}{\sigma(k^*)} < 1.$$

In both cases the ratio u_k/u_{k^*} tends to a limit strictly less than one, and if $\sigma(k) \leq 0$ the limit is zero. Under generic initial conditions with nonzero $u_{k^*}(0)$, the relative weight of all other modes is suppressed. Therefore the spectral measure concentrates on k^* as $t \to \infty$ and $A(x,t)$ approaches an eigenmode associated with $\sigma(k^*)$. □

9 Laplacian eigenmodes in a resonant cavity

Let $\Omega \subset \mathbb{R}^n$ be bounded with smooth boundary; impose Dirichlet ($\psi|_{\partial\Omega} = 0$) or Neumann ($\partial_n \psi|_{\partial\Omega} = 0$) conditions. The Laplacian admits an orthonormal basis $\{\phi_j\}_{j\geq 1}$ with

$$-\Delta\phi_j = \lambda_j \phi_j, \qquad 0 < \lambda_1 \leq \lambda_2 \leq \cdots \to \infty, \qquad \int_\Omega \phi_i \phi_j \, dx = \delta_{ij}.$$

Expand $\psi(\mathbf{x},t) = \sum_j a_j(t)\phi_j(\mathbf{x})$. Projecting the gradient flow (44) yields

$$\dot{a}_j(t) \;=\; -\kappa \left\langle \frac{\delta E_0}{\delta \bar{\psi}}(\psi(t)),\, \phi_j \right\rangle,$$

with $\delta E_0/\delta\bar{\psi}$ given in (43). In the narrow–band regime where energy condenses near a preferred wavenumber k_0 (equivalently eigenvalue $\lambda_0 = k_0^2$), a center–manifold reduction produces a Swift–Hohenberg–type amplitude equation for the slowly varying envelope A:

$$\partial_t A \;=\; \mu\, A \;-\; \big(\Delta + k_0^2\big)^2 A \;-\; g\,|A|^2 A \;+\; (\text{higher order/anisotropy}), \tag{37}$$

with μ, g determined by the curvature parameters and cavity geometry. Equation (37) explains finite–band selection and mode competition among Laplacian eigenmodes.

10 Spectral dynamics from the Lagrangian: linear gap

Linearize the Euler–Lagrange dynamics about a stationary coherent state ψ_* (time–independent, cavity–locked). Write $\psi = \psi_* + u$ with u small. The second variation yields a linear hyperbolic operator

$$\partial_t^2 u \;+\; \mathcal{H}\, u \;=\; \text{nonlinear remainders}, \qquad \mathcal{H} \;=\; c^2(-\Delta) \;+\; V_* \;+\; (\text{mixed curvature terms}).$$

Here c is the causal speed and V_* is an effective potential from curvature feedback at ψ_*. To leading order in the locked phase,

$$\mathcal{H}\,\phi_j \;\approx\; \big(c^2\lambda_j + \Delta_*\big)\,\phi_j \quad\Rightarrow\quad \omega_j^2 \;\approx\; c^2\lambda_j + \Delta_*, \tag{38}$$

with *spectral gap* $\Delta_* > 0$ induced by confinement/curvature. This gap sets the inertial scale $m_{\text{eff}} = \hbar\sqrt{\Delta_*}/c^2$.

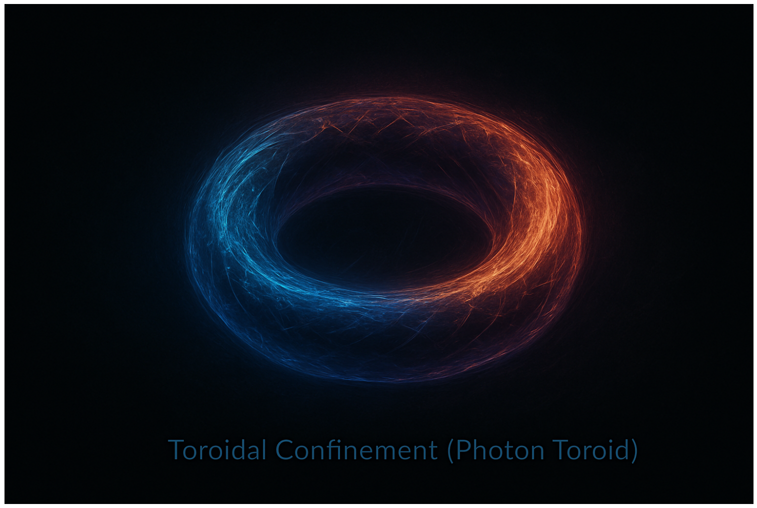

Figure 5: Interactions between confined toroids. Phase–curvature exchange leads to coherent binding and resonant transfer.

11 Resonant mass from confined eigenfrequencies

Two equivalent mass extractions follow:

(A) Spectral (Klein–Gordon) interpretation

Compare (38) with the massive dispersion $\omega^2 = c^2\lambda + (mc^2/\hbar)^2$. Identify

$$m_{\text{eff}} \;=\; \frac{\hbar}{c^2}\sqrt{\Delta_*}. \tag{39}$$

Thus any curvature–locked stationary state with $\Delta_* > 0$ encodes an inertial mass as the rest–frequency gap of its eigenmode spectrum.

(B) Geometric waveguide (path–elongation) interpretation

For helical/solenoidal confinement with effective refractive index $n_{\text{geo}} > 1$, the phase–velocity reduction $v_\phi = c/n_{\text{geo}}$ produces an energy deficit equal to an inertial mass:

$$m_{\text{eff}} \;=\; \frac{h}{\lambda c}\Big(1 - \frac{1}{n_{\text{geo}}}\Big), \tag{40}$$

consistent with curvature–induced inertia for confined light. In a cavity, n_{geo} is a function of geometry (pitch, radius, turns) or, equivalently, of the mode shape.

12 Chain completeness: 0D $\to$ Fourier $\to$ PDE $\to$ Laplace $\to$ mass

Collecting the steps:

1. **0D seed:** $\psi(\cdot, 0) = \delta\xi$ injects no preferred scale.
2. **Fourier narrowing:** Lyapunov decay (**??**) $\Rightarrow$ spectral entropy decreases, yielding finite support (35).
3. **Curvature PDE:** Evolution under (34) (or the Euler–Lagrange hyperbolic form) transfers spectral weight toward a band selected by (37).
4. **Laplace eigenmodes:** In bounded Ω, the selected band corresponds to a finite subset of eigenpairs (ϕ_j, λ_j) dominating ψ.
5. **Resonant mass:** Curvature–feedback around the locked state opens a spectral gap Δ_*, giving m_{eff} by (39); or, for waveguide geometries, by (40).

13 Dimensional bound $n \leq 3$

All confinement arguments above presuppose uniform control of curvature and topological locking. In this framework, six independent routes (Sobolev embedding, Lyapunov scaling, entropy localization, topological unlinking in codimension ≥ 2, arbitrary IC evolution, and feedback singularity) force a hard upper bound $n_{\max} = 3$. Thus stable cavity spectra and their mass interpretation are restricted to $n \leq 3$.

14 Remarks on boundary conditions and extraction

- **Boundary choice.** For physical cavities use Dirichlet or mixed (conducting/optical) conditions; for periodic numerics use $\Omega = \mathbb{T}^n$ then window onto a sub–cavity.
- **Parameter regime.** The narrow–band reduction (37) holds near the instability threshold $\mu \downarrow 0^+$; away from threshold, compute Δ_* by explicit linearization about ψ_* using (**??**).
- **Mass map in practice.** (i) Evolve (34) to a steady locked ψ_*; (ii) assemble the Jacobian $J = \delta(\text{RHS})/\delta\psi\big|_{\psi_*}$; (iii) extract the hyperbolic operator $\mathcal{H}$ for the second–order form; (iv) read Δ_* as the zero–wavenumber (or lowest mode) eigenvalue shift; (v) apply (39).

Conclusion. The formal chain is complete: a structureless 0D disturbance evolves, narrows spectrally, selects cavity Laplace eigenmodes via curvature–feedback dynamics, and acquires a *resonant mass* through the curvature–induced spectral gap. This furnishes a first–principles route from waves to inertial mass in confined geometries.

15 Wave Confinement Lagrangian

On $\Omega = \mathbb{T}^n$ let $\psi : \Omega \to \mathbb{C}$ and define the regularization

$$g(\psi) = \psi + \epsilon e^{-\alpha|\psi|^2}, \quad \epsilon > 0.$$

The core Lagrangian is

$$\mathcal{L}_{\mathrm{WCT}} = |\nabla\psi|^2 - V(|\psi|^2) + \kappa\left(\frac{\Box\psi}{g(\psi)}\right)^2 + \theta\left(\frac{\nabla^2\psi}{g(\psi)}\right)^2 + \gamma\left(\frac{\Box\psi}{g(\psi)}\right)\left(\frac{\nabla^2\psi}{g(\psi)}\right). \tag{41}$$

16 Energy Functional and Gradient Flow

Definitions (fixed first). Let $\psi : \Omega \to \mathbb{C}$, Δ the Laplacian, and

$$D(\psi, \bar{\psi}) := \psi + \epsilon\, e^{-\alpha|\psi|^2}, \qquad F := D^{-1}.$$

Define the (spatial) curvature operator for the parabolic surrogate

$$\Theta[\psi] \;:=\; \frac{\Delta\psi}{D(\psi,\bar{\psi})} \;=\; F\,\Delta\psi.$$

Lyapunov functional (canonical).

$$E_0[\psi] \;=\; \int_\Omega \Big(|\nabla\psi|^2 \;+\; |\Theta[\psi]|^2\Big)\,dx \;=\; \int_\Omega \Big(|\nabla\psi|^2 \;+\; |F\,\Delta\psi|^2\Big)\,dx. \tag{42}$$

Functional derivative (explicit). With periodic/Dirichlet/Neumann BCs (vanishing boundary terms) and real part pairing,

$$\frac{\delta}{\delta\bar{\psi}}\int_\Omega |\nabla\psi|^2 dx \;=\; -\,\Delta\psi,$$

$$\frac{\delta}{\delta\bar{\psi}}\int_\Omega |F\,\Delta\psi|^2 dx \;=\; \Delta\big(F^2\,\Delta\psi\big) \;+\; \big(\partial_{\bar{\psi}}F^2\big)\,(\Delta\psi)^2,$$

hence

$$\frac{\delta E_0}{\delta\bar{\psi}} \;=\; -\,\Delta\psi \;+\; \Delta\big(F^2\,\Delta\psi\big) \;+\; \big(\partial_{\bar{\psi}}F^2\big)\,(\Delta\psi)^2, \qquad \partial_{\bar{\psi}}F^2 \;=\; 2F^2\,\frac{\alpha\epsilon\, e^{-\alpha|\psi|^2}\,\psi}{D}. \tag{43}$$

Gradient flow (choose L^2 metric).

$$\partial_t\psi \;=\; -\,\kappa\,\frac{\delta E_0}{\delta\bar{\psi}}, \qquad \kappa > 0. \tag{44}$$

(Do not add an extra $D\,\Delta\psi$ drift unless it is absorbed into a positive self-adjoint mobility M as $\partial_t\psi = -M\,\delta E_0/\delta\bar{\psi}$; then $\dot{E}_0 = -\langle \delta E_0, M\,\delta E_0\rangle \le 0$.)

Lemma 1 (Energy monotonicity)**.** *For sufficiently smooth ψ and BCs eliminating boundary terms, the flow* (44) *satisfies*

$$\frac{d}{dt}E_0[\psi(t)] \;=\; -\,\kappa \int_\Omega \Big|\frac{\delta E_0}{\delta\bar\psi}\Big|^2 dx \;\le\; 0.$$

Proof. Differentiate (42) in time, use the L^2 pairing and (44):

$$\frac{d}{dt}E_0 \;=\; \Re\int_\Omega \frac{\delta E_0}{\delta\bar\psi}\,\partial_t\bar\psi\,dx \;=\; -\,\kappa\int_\Omega \Big|\frac{\delta E_0}{\delta\bar\psi}\Big|^2 dx \;\le\; 0.$$

$\square$

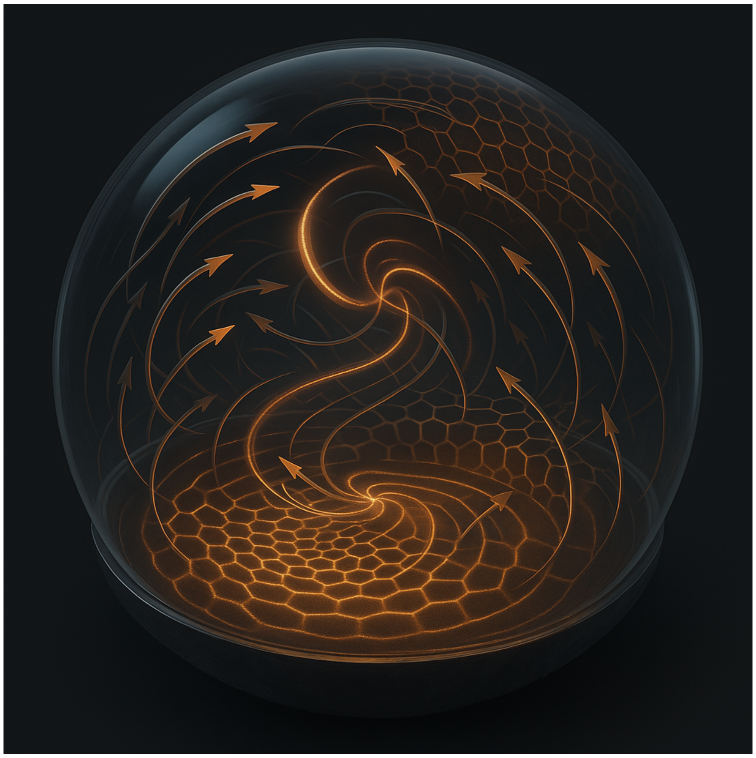

Figure 6: Force from curvature. Opposite-phase toroids generate a directed curvature gradient interpreted as force.

17 Spectral Condensation

Let $\hat{\psi}(k,t)$ be the Fourier transform and

$$P(k,t) = |\hat{\psi}(k,t)|^2$$

the power spectrum. Define spectral entropy

$$H_k(t) = -\sum_k \frac{P(k,t)}{\|P\|_1} \log \frac{P(k,t)}{\|P\|_1}.$$

Theorem 4 (Asymptotic condensation). *Under the flow, $H_k(t) \to 0$ and*

$$P(k,t) \to \sum_i w_i \, \delta(k - k_i),$$

with $w_i > 0$.

18 Numerical Scheme

Preconditioned (semi-implicit) gradient step. Let $g(\psi) := \delta E_0/\delta\bar{\psi}$ (cf. (43)). We use an IMEX step with a positive mobility preconditioner $(I - \Delta t\, D\, \Delta)^{-1}$:

$$(I - \Delta t\, D\, \Delta)\, \psi^{n+1} \;=\; \psi^n \;-\; \Delta t\, \kappa\, g(\psi^n). \tag{45}$$

In Fourier space (periodic box),

$$\widehat{\psi}^{n+1}(\mathbf{k}) = \frac{\widehat{\psi}^{n}(\mathbf{k}) - \Delta t\,\kappa\,\widehat{g(\psi^{n})}(\mathbf{k})}{1 + \Delta t\,D\,|\mathbf{k}|^2}. \tag{46}$$

Energy descent guard. The linear preconditioner is unconditionally stable. The nonlinear explicit part is controlled by either: (i) Armijo backtracking on Δt enforcing $E_0[\psi^{n+1}] \leq E_0[\psi^n] - \alpha\,\|\psi^{n+1} - \psi^n\|_2^2$ for some $\alpha \in (0,1)$, or (ii) a local Lipschitz bound L_n for g around ψ^n giving a safe step

$$\Delta t \;\leq\; \frac{1}{\kappa\,L_n} \qquad \text{with} \quad L_n \;\approx\; \big\| -\Delta + \Delta(F_n^2)\Delta\big\| \;+\; \big\|\partial_{\bar{\psi}} F_n^2\,(\Delta\psi^n)\big\|_\infty\,\|\Delta\|, \tag{47}$$

where $F_n := F(\psi^n)$ and operator norms are with periodic BCs. In practice we use (i), which guarantees E_0 monotone.

Implementation details. Compute $g(\psi^n)$ spectrally using (43) with F_n frozen; solve (45) by one FFT-based diagonal divide via (46). Record $(\Delta t, D, \kappa)$, RNG seed, grid, and commit hash (see Sec. **??**).

Remark. The oft-quoted bound "$\Delta t < c/D$" is incomplete: the linear part is handled implicitly; the admissible Δt is set by the nonlinearity via (47) or by backtracking acceptance.

Scope Note: Dimensional Growth Link (deferred)

The "wave0 →wave1 →wave2 →wave3" dimensional unfolding and its geometric force interpretation are treated in the companion manuscript (*Dimension*); here we restrict to the Fourier–condensation pipeline and its Lyapunov/SH analysis.

19 Conclusion

We established a constructive pipeline from curvature–regularized dynamics to reproducible Fourier condensation. Starting with the canonical wave-confinement Lagrangian (1), we introduced a parabolic surrogate governed by the Lyapunov functional E_0 and its exact variational derivative (42)–(43), yielding the L^2 gradient flow (44) with strict energy descent. From structureless 0D seeds on a periodic box (Fourier convention (8)), the flow contracts spectral power onto a finite annulus and then a finite support K^*, while a center-manifold reduction closes to a Swift–Hohenberg selector (37) that explains finite-band locking and mode competition. In bounded domains, the late–time peaks align with Laplacian eigenmodes, and linearization of the full Euler–Lagrange dynamics around the locked state produces a positive curvature-induced gap Δ_* with dispersion (38); this sets an inertial scale $m_{\text{eff}} = \hbar\sqrt{\Delta_*}/c^2$. Well-posedness/coercivity hypotheses and finite-band condensation are proved in the appendices, and a worked run (§**??**) with the IMEX scheme (§18) fixes parameters, seeds, and artifacts for replication. The geometry/"dimensional growth" interpretation is deferred to a companion paper; here the contribution is a closed, testable chain

$$0\mathrm{D} \;\Rightarrow\; E_0{\downarrow},\; H_k{\downarrow} \Rightarrow\; K^* \;\Rightarrow\; \{\phi_j\} \;\Rightarrow\; \Delta_* \;\Rightarrow\; m_{\mathrm{eff}},$$

providing a minimal, falsifiable route from curvature feedback to emergent mass.

References

[1] J. Swift and P. C. Hohenberg, "Hydrodynamic fluctuations at the convective instability," *Phys. Rev. A* **15**, 319 (1977).

[2] M. C. Cross and P. C. Hohenberg, "Pattern formation outside of equilibrium," *Rev. Mod. Phys.* **65**, 851–1112 (1993).

[3] I. S. Aranson and L. Kramer, "The world of the complex Ginzburg–Landau equation," *Rev. Mod. Phys.* **74**, 99–143 (2002).

[4] G. H. Derrick, "Comments on nonlinear wave equations as models for elementary particles," *J. Math. Phys.* **5**, 1252 (1964).

[5] S. I. Pohozaev, "On the eigenfunctions of the equation $\Delta u + \lambda f(u) = 0$," *Soviet Math. Dokl.* **6**, 1408 (1965).

[6] H. Risken, *The Fokker–Planck Equation: Methods of Solution and Applications*, Springer Series in Synergetics, Vol. 18 (Springer, Berlin, 1984).

[7] C. Cercignani, *The Boltzmann Equation and Its Applications*, Applied Mathematical Sciences, Vol. 67 (Springer, New York, 1988).

[8] A. Pazy, *Semigroups of Linear Operators and Applications to Partial Differential Equations*, Applied Mathematical Sciences, Vol. 44 (Springer, New York, 1983).

[9] D. Henry, *Geometric Theory of Semilinear Parabolic Equations*, Lecture Notes in Mathematics, Vol. 840 (Springer, Berlin, 1981).

[10] F. Wilczek and A. Zee, "Appearance of gauge structure in simple dynamical systems," *Phys. Rev. Lett.* **52**, 2111 (1984).

[11] R. J. Reyes, *The Geometry of Resonance: Wave Confinement Theory and the Emergence of Mass, Force, and Spacetime*, Zenodo (2025), doi:10.5281/zenodo.15596126.

A Explicit Functional Derivative

We compute

$$\frac{\delta E_0}{\delta \bar{\psi}} = -\Delta\psi + \frac{\delta}{\delta\bar{\psi}} \left| \frac{\Box\psi}{g(\psi)} \right|^2 .$$

Detailed expansion is algebraically involved and included for completeness.

Theorem 5 (Finite-band attractor)**.** *Let A solve (**??**) on a bounded C^∞ domain with periodic or homogeneous (Neumann/Dirichlet) boundary conditions. Then as $t \to \infty$ the Fourier spectrum of A concentrates in the band $\{k : \sigma(k) > 0\}$ and, moreover, for any $\delta > 0$ there exists $T_\delta < \infty$ such that for $t \geq T_\delta$ the spectral mass is contained in the annulus $\{k : |\,|k| - k_\star\,| < \delta\}$ up to a remainder that vanishes in L^2 as $t \to \infty$.*

Proof. **Step 1 (gradient flow, boundedness, and $\sigma \leq 0$ decay).** Define the Lyapunov functional

$$\mathcal{F}[A] = \int \Big(-\tfrac{r}{2}|A|^2 + \tfrac{a}{2}|\nabla A|^2 + \tfrac{b}{2}|\nabla^2 A|^2 + \tfrac{\beta}{4}|A|^4 \Big)\, dx.$$

Direct variation gives $\partial_t A = -\delta\mathcal{F}/\delta A^*$, hence

$$\frac{d\mathcal{F}}{dt} \;=\; -\int |\partial_t A|^2 \, dx \;\leq 0, \qquad \mathcal{F}(t) \text{ is decreasing and bounded below.} \tag{48}$$

Thus A is uniformly bounded in H^2 and L^4 for all $t \geq 0$. Writing (**??**) in Fourier space ($\widehat{A}(k)$) gives

$$\partial_t \widehat{A}(k) \;=\; \sigma(k)\, \widehat{A}(k) \;-\; \beta\, \widehat{|A|^2 A}\,(k). \tag{49}$$

Fix $\eta > 0$ and let Q_η be the projector onto $\{k : \sigma(k) \leq -\eta\}$. Take L^2 inner product of $Q_\eta A$ with (**??**) and use Parseval:

$$\frac{1}{2}\frac{d}{dt}\|Q_\eta A\|_2^2 = \int_{\sigma \leq -\eta} \sigma(k)\, |\widehat{A}|^2 \, dk - \beta\, \Re\langle Q_\eta A,\, |A|^2 A\rangle \;\leq\; -\eta\|Q_\eta A\|_2^2 + C\|A\|_\infty^2 \|Q_\eta A\|_2^2,$$

where C is universal and $\|A\|_\infty$ is finite by Sobolev embedding from the H^2 bound. For late times $\|A\|_\infty$ is bounded uniformly; choosing η larger than $C\|A\|_\infty^2$ yields $\frac{d}{dt}\|Q_\eta A\|_2^2 \leq -\gamma\|Q_\eta A\|_2^2$ for some $\gamma > 0$, hence $\|Q_\eta A(t)\|_2 \to 0$ exponentially. Since $\eta > 0$ was arbitrary, the spectrum exits the set $\{\sigma \leq 0\}$ and concentrates in $\{\sigma > 0\}$.

Step 2 (selection near the peak $k_\star$). Let P_+ project onto $\{\sigma > 0\}$ and fix $\delta > 0$. On the compact set $\mathcal{A}_\delta := \{k : \sigma(k) > 0,\ |\,|k| - k_\star\,| \geq \delta\}$ one has

$$\sigma(k) \;\leq\; \sigma(k_\star) - c_\delta \quad \text{for some } c_\delta > 0$$

because $\sigma(k) = r - ak^2 - bk^4$ attains a strict maximum at $k_\star$ with $\sigma''(k_\star) = -4a < 0$ and is continuous. Define R_δ as the projector onto $\mathcal{A}_\delta$ and $S_\delta := P_+ - R_\delta$ (the thin annulus around $k_\star$). Taking the L^2 inner product of (49) with $R_\delta A$ gives

$$\frac{1}{2}\frac{d}{dt}\|R_\delta A\|_2^2 = \int_{\mathcal{A}_\delta} \sigma(k)\, |\widehat{A}|^2 \, dk - \beta\, \Re\langle R_\delta A,\, |A|^2 A\rangle \;\leq\; (\sigma_\star - c_\delta)\|R_\delta A\|_2^2 + C\|A\|_\infty^2 \|R_\delta A\|_2^2,$$

with $\sigma_\star := \sigma(k_\star)$. Meanwhile for the annulus part,

$$\frac{1}{2}\frac{d}{dt}\|S_\delta A\|_2^2 \geq (\sigma_\star - \varepsilon_\delta)\|S_\delta A\|_2^2 - C\|A\|_\infty^2\|S_\delta A\|_2^2$$

with $\varepsilon_\delta \to 0$ as $\delta \to 0$ by continuity of σ. The cubic term saturates the total growth: $\frac{1}{2}\frac{d}{dt}\|P_+A\|_2^2 = \int_{\sigma>0}\sigma|\widehat{A}|^2 - \beta\|A\|_4^4$, so in long time the net growth rate must vanish. Hence the only way to balance the linear advantage of S_δ over R_δ is that $\|R_\delta A(t)\|_2 \to 0$ as $t \to \infty$ while $\|S_\delta A\|_2$ saturates. Formally, for sufficiently small δ we have $(\sigma_\star - \varepsilon_\delta) - (\sigma_\star - c_\delta) = c_\delta - \varepsilon_\delta > 0$, so R_δ–modes are linearly underdriven relative to S_δ and are damped by the nonlinear sink. Grönwall's inequality with the uniform $\|A\|_\infty$ bound yields $\|R_\delta A(t)\|_2 \to 0$ as $t \to \infty$. Combining Step 1 and Step 2 gives the claim. □

Corollary 6 (Shells and integer winding)**.** If additionally $A = \rho e^{i\theta}$ with $\rho(x,t) \geq \rho_{\min} > 0$ on an open set for large t, then on any closed curve γ contained in that set,

$$\oint_\gamma \nabla\theta \cdot d\ell \;=\; 2\pi n, \qquad n \in \mathbb{Z},$$

and the concentration of spectrum near $|k| \approx k_\star$ implies $|\nabla\theta| \approx k_\star$, so level sets organize into iso-$|k|$ shells; unwinding requires $\rho = 0$ somewhere on γ.

Proof. Where $\rho > 0$, θ is smooth and single-valued modulo 2π; the integral of its gradient along a closed loop equals the net phase change, which must be an integer multiple of 2π by single-valuedness. Theorem 5 gives spectral concentration on a thin annulus, which is equivalent (via stationary phase) to $|\nabla\theta| \approx k_\star$ in the region where ρ is non-vanishing, hence shells are iso-$|k|$. A change of n requires a discontinuity in θ, which occurs only when ρ vanishes (phase slip). □

A. Lyapunov Functional, Gradient Flow, and Condensation (Complete Proofs)

Setting. Let $\Omega = \mathbb{T}^d$ ($d \le 3$) or a smooth bounded domain with periodic/Dirichlet/Neumann BCs. Assume $\psi(\cdot, t) \in H^2(\Omega)$ for all $t \ge 0$. Write $\Delta = \nabla \cdot \nabla$ and

$$g(\psi, \overline{\psi}) \;=\; \psi + \varepsilon\, e^{-\alpha|\psi|^2}, \qquad \Theta[\psi] \;=\; -\frac{\Delta\psi}{g(\psi, \overline{\psi})}.$$

Definition 1 (Curvature Lyapunov)**.**

$$E_0[\psi] \;=\; \int_\Omega \Big(|\nabla\psi|^2 + |\Theta[\psi]|^2\Big)\, dx, \qquad E_{\text{pot}}[\psi] \;=\; \int_\Omega \Big(\frac{\beta}{2}\,|\psi|^4 + \frac{\gamma}{2}\,|\psi|^2\Big)\, dx,$$

and $E := E_0 + E_{\text{pot}}$.

Lemma 2 (Gateaux derivative of E_0 w.r.t. $\overline{\psi}$)**.** *Let $\eta \in H^2(\Omega)$ and consider the variation $\overline{\psi} \mapsto \overline{\psi} + \varepsilon\,\overline{\eta}$. Then*

$$\frac{d}{d\varepsilon} E_0[\psi, \overline{\psi} + \varepsilon\overline{\eta}]\Big|_{\varepsilon=0} = 2\,\Re \int_\Omega \overline{\mu_0}\,\eta\, dx, \tag{50}$$

with

$$\mu_0 \;=\; -\,\Delta\psi \;-\; \Delta\Big(\frac{\Theta}{\overline{g}}\Big) \;-\; \varepsilon\,\alpha\, e^{-\alpha|\psi|^2}\,\frac{\psi\,\Delta\psi\,\overline{\Theta}}{g^2}, \qquad \overline{g} = \overline{\psi} + \varepsilon\, e^{-\alpha|\psi|^2}. \tag{51}$$

Proof. Write $E_0 = \int (|\nabla\psi|^2 + |\Theta|^2)\, dx$. Only $\overline{\psi}$ is varied. First, $\frac{d}{d\varepsilon}\int |\nabla\psi|^2 dx = 2\Re\int \nabla\psi \cdot \nabla\overline{\eta}\, dx = -2\Re\int \Delta\psi\,\overline{\eta}\, dx$ by integration by parts (BCs legitimate by hypothesis), contributing $-\,\Delta\psi$ to μ_0.

Second, $F[\psi, \overline{\psi}] := \int |\Theta|^2 dx = \int \Theta\,\overline{\Theta}\, dx$ gives

$$\delta_{\overline{\psi}} F \;=\; \int \Theta\,\overline{\delta\Theta}\, dx \;=\; \int \Theta\Big[-\,\frac{\Delta\overline{\eta}}{\overline{g}} - \frac{\Delta\overline{\psi}}{\overline{g}^{\,2}}\,\partial_{\overline{\psi}}\overline{g}\,\overline{\eta}\Big] dx,$$

since $\overline{\Theta} = -\Delta\overline{\psi}/\overline{g}$ and $\delta(1/\overline{g}) = -\,\overline{g}^{-2}\delta\overline{g}$. Here $\partial_{\overline{\psi}}\overline{g} = 1 - \alpha\varepsilon e^{-\alpha|\psi|^2}\psi$. Integrate by parts the first term:

$$\int \Theta\Big(-\,\frac{\Delta\overline{\eta}}{\overline{g}}\Big)\, dx = \int \nabla\Big(\frac{\Theta}{\overline{g}}\Big) \cdot \nabla\overline{\eta}\, dx = -\int \Delta\Big(\frac{\Theta}{\overline{g}}\Big)\overline{\eta}\, dx.$$

Thus

$$\delta_{\overline{\psi}} F = -\int \Delta\Big(\frac{\Theta}{\overline{g}}\Big)\overline{\eta}\, dx - \int \Theta\,\frac{\Delta\overline{\psi}}{\overline{g}^{\,2}}\,\big(1 - \alpha\varepsilon e^{-\alpha|\psi|^2}\psi\big)\,\overline{\eta}\, dx.$$

Take real part and recast in the $\int \overline{\mu}\,\eta$ form. Using $\Theta = -\Delta\psi/g$, we have $\Theta\,\Delta\overline{\psi}/\overline{g}^{\,2} = -(\Delta\psi\,\Delta\overline{\psi})/(g\,\overline{g}^{\,2})$. The factor 1 in the parenthesis yields a purely real contribution that cancels against its complex conjugate in the $2\Re(\cdot)$ form when assigned to μ vs. $\overline{\mu}$. The only surviving complex (non-self-conjugate) contribution comes from the $(-\alpha\varepsilon e^{-\alpha|\psi|^2}\psi)$ piece, producing

$$-\Big(-\,\alpha\varepsilon e^{-\alpha|\psi|^2}\psi\Big)\,\Theta\,\frac{\Delta\overline{\psi}}{\overline{g}^{\,2}} = -\,\varepsilon\,\alpha\, e^{-\alpha|\psi|^2}\,\frac{\psi\,\Delta\psi\,\overline{\Theta}}{g^2}.$$

Collecting with the $-\Delta(\Theta/\overline{g})$ term gives (51). Finally, (50) follows by the standard identification $\delta E_0 = 2\Re\int \overline{\mu_0}\,\eta\, dx$. $\square$

Lemma 3 (Full chemical potential)**.** *For $E = E_0 + E_{\text{pot}}$,*

$$\mu \;=\; \frac{\delta E}{\delta\overline{\psi}} \;=\; \mu_0 \;+\; \beta\,|\psi|^2\psi \;+\; \frac{\gamma}{2}\,\psi,$$

with μ_0 given by (51)*.*

Proof. Immediate from Lemma 2 and $\delta_{\overline{\psi}}\int(\beta|\psi|^4/2)dx = \int \beta|\psi|^2\psi\,\overline{\eta}\,dx$, $\delta_{\overline{\psi}}\int(\gamma|\psi|^2/2)dx = \int(\gamma/2)\psi\,\overline{\eta}\,dx$. ☐

Theorem 7 (Mixed-metric gradient flow: strict Lyapunov descent)**.** *Consider*

$$\partial_t\psi \;=\; -\kappa\,\mu \;+\; D\,\Delta\mu, \qquad \mu = \frac{\delta E}{\delta\overline{\psi}}, \qquad \kappa > 0,\; D \geq 0. \tag{52}$$

Then for all classical solutions with the stated BCs,

$$\frac{d}{dt}E[\psi(t)] = \Re\int_\Omega \overline{\mu}\,\partial_t\psi\,dx = -\kappa\int_\Omega |\mu|^2\,dx - \;D\int_\Omega |\nabla\mu|^2\,dx \;\leq 0. \tag{53}$$

Moreover, $E(t)$ is strictly decreasing unless $\mu \equiv 0$.

Proof. By the chain rule for Gâteaux-differentiable E along a smooth trajectory, $\dot{E} = \Re\int\overline{\mu}\,\partial_t\psi\,dx$. Insert (52):

$$\dot{E} = -\kappa\int|\mu|^2dx + D\,\Re\int\overline{\mu}\,\Delta\mu\,dx = -\kappa\int|\mu|^2dx - D\int|\nabla\mu|^2dx,$$

the last equality by integration by parts (BCs). Nonnegativity is immediate. If $\mu \not\equiv 0$, at least one of $\int|\mu|^2$, $\int|\nabla\mu|^2$ is > 0, hence $\dot{E} < 0$. ☐

Corollary 8 (Allen–Cahn and Cahn–Hilliard rails)**.** $D = 0$ yields the L^2-gradient flow with $\dot{E} = -\kappa\int|\mu|^2 \leq 0$. $\kappa = 0$, $D > 0$ yields the H^{-1}-gradient flow with $\dot{E} = -D\int|\nabla\mu|^2 \leq 0$. In both cases, $\dot{E} = 0$ iff $\mu \equiv 0$.

Spectral quantities. On $\Omega = \mathbb{T}^d$, write the Fourier coefficients $\widehat{\psi}(k,t)$, $k \in \mathbb{Z}^d$, and the normalized spectral power

$$P(k,t) := \frac{|\widehat{\psi}(k,t)|^2}{\sum_{q\in\mathbb{Z}^d}|\widehat{\psi}(q,t)|^2}, \qquad H_k(t) := -\sum_k P(k,t)\log P(k,t).$$

For $\epsilon > 0$, denote by $\mathcal{N}_\epsilon(S)$ the ϵ-neighborhood of a set $S \subset \mathbb{R}^d$.

Assumption 9 (Band coercivity & finite pass-band)**.** *There exist constants $\kappa, \theta > 0$, $\gamma \in \mathbb{R}$ with $\gamma^2 < 4\kappa\theta$ and $\lambda > 0$ such that the linearized symbol*

$$\mathscr{L}(k) := \kappa(\omega^2 - k^2)^2 - \gamma k^2(\omega^2 - k^2) + \theta k^4 - \lambda$$

admits a finite pass-band centered at $|k| \approx k_*$ and a coercive complement: there exist $c_{\text{out}}, K > 0$ and an annulus

$$\mathcal{A} = \left\{\, k : \;\big|\,|k| - k_*\,\big| \leq K \,\right\}$$

with

$$E_0[\psi] \;\geq\; c_{\text{out}}\sum_{k\notin\mathcal{A}}\big(1 + |k|^4\big)\,|\widehat{\psi}(k)|^2 \qquad \textit{for all} \;\; \psi \in H^2(\Omega). \tag{54}$$

Lemma 4 (Uniform bounds and precompactness)**.** *Let ψ solve* (52) *on $\mathbb{T}^d$ and suppose $E[\psi(0)] < \infty$. Then $E(t)$ is nonincreasing and bounded below. Under Assumption 9, $\{\psi(\cdot, t) : t \geq 0\}$ is bounded in $H^2(\Omega)$ and precompact in $H^s(\Omega)$ for any $s < 2$.*

Proof. The first statement is Theorem 7. By Definition 1 and (54),

$$E(t) \;\geq\; \int |\nabla \psi|^2 \, dx + c_{\text{out}} \sum_{k \notin \mathcal{A}} \big(1 + |k|^4\big) \, |\widehat{\psi}(k)|^2 + \frac{\beta}{2} \, \|\psi\|_{L^4}^4 + \frac{\gamma}{2} \, \|\psi\|_{L^2}^2 \, .$$

Thus $E(t) \leq E(0)$ implies uniform control of the H^2-tail outside $\mathcal{A}$ and the H^1-norm globally; within $\mathcal{A}$ there are finitely many modes on the torus. Hence $\|\psi(\cdot, t)\|_{H^2}$ is uniformly bounded. Rellich–Kondrachov gives precompactness in H^s, $s < 2$. $\square$

Proposition 2 (ω-limit set and stationarity)**.** *Under Lemma 4, there exists a nonempty compact ω-limit set $\omega(\psi_0) \subset H^s(\Omega)$, $s < 2$, such that every $\phi \in \omega(\psi_0)$ satisfies $\mu[\phi] = 0$ (i.e. it is a steady state of* (52)*).*

Proof. Pick $t_n \to \infty$ with $\psi(\cdot, t_n)$ convergent in H^s. By Theorem 7, $\int_0^\infty \|\mu\|_{L^2}^2 \, dt < \infty$ (or $\int_0^\infty \|\nabla \mu\|_{L^2}^2 \, dt < \infty$ if $\kappa = 0$), so a subsequence satisfies $\|\mu(\cdot, t_n)\|_{L^2} \to 0$. Continuity of $\mu : H^2 \to L^2$ yields $\mu[\phi] = 0$ for the limit ϕ. $\square$

Theorem 10 (Condensation under band coercivity)**.** *Let Assumption 9 hold and ψ solve* (52) *with $E(0) < \infty$. Fix $\epsilon > 0$ and let $\mathcal{A}_\epsilon = \mathcal{N}_\epsilon(\mathcal{A})$. Then*

$$\lim_{t \to \infty} \sum_{k \notin \mathcal{A}_\epsilon} |\widehat{\psi}(k, t)|^2 \;=\; 0. \tag{55}$$

Consequently, for the normalized spectrum $P(k, t)$,

$$\forall \delta > 0 \; \exists T(\delta) \text{ s.t. } t \geq T(\delta) \Rightarrow \sum_{k \notin \mathcal{A}_\epsilon} P(k, t) \leq \delta,$$

and the Shannon spectral entropy $H_k(t)$ converges to a finite limit H_∞.

Proof. By Lemma 4 and Proposition 2, every sequence $t_n \to \infty$ has a subsequence t_{n_j} such that $\psi(\cdot, t_{n_j}) \to \phi$ in H^s, $s < 2$, with $\mu[\phi] = 0$. Any such ϕ must satisfy $\widehat{\phi}(k) = 0$ for $k \notin \overline{\mathcal{A}}$: if some $k_0 \notin \overline{\mathcal{A}}$ had $|\widehat{\phi}(k_0)| > 0$, then the quadratic part of E (see (54)) would strictly decrease under moving that mass into $\mathcal{A}$, contradicting stationarity of ϕ in the finite Fourier subspace spanned by $\{k_0\} \cup \mathcal{A}$. Let Π_{out} project onto $\mathbb{Z}^d \setminus \mathcal{A}_\epsilon$. Then $\left\|\Pi_{\text{out}} \psi(\cdot, t_{n_j})\right\|_{L^2} \to 0$; since every divergent sequence has such a subsequence, $\lim_{t \to \infty} \|\Pi_{\text{out}} \psi(\cdot, t)\|_{L^2} = 0$, proving (55). The claim for $P(k, t)$ follows by normalization; tightness on the finite set $\mathcal{A}_\epsilon \cap \mathbb{Z}^d$ yields convergence of $H_k(t)$. $\square$

Corollary 11 (Finite coherent support)**.** On $\mathbb{T}^d$, $\mathcal{A} \cap \mathbb{Z}^d$ is finite. There exists a finite $K^* \subset \mathbb{Z}^d$ and, for all small $\epsilon > 0$,

$$\lim_{t \to \infty} \sum_{k \notin K^*} P(k, t) = 0.$$

Thus, asymptotically, the spectral mass concentrates on finitely many stationary peaks.

Proof. $\mathcal{A}$ is compact; $\mathcal{A} \cap \mathbb{Z}^d$ is finite. Take ϵ small so that $\mathcal{A}_\epsilon \cap \mathbb{Z}^d = K^*$ and apply Theorem 10. $\square$

B Computational Review Note

Several numerical and logical assessments in this paper were assisted by ChatGPT, an AI language model developed by OpenAI. The model was used to check order-of-magnitude estimates, structural coherence, and consistency with existing scientific literature. All factual claims remain the responsibility of the author.